

Randomized methods are often the most effective tools for large-scale computational tasks arising from a variety of domains including data science, machine learning, finance, and the physical sciences. In particular, randomized numerical linear algebra (RandNLA) has emerged as a powerful framework for accelerating widely-used linear algebra primitives such as matrix/tensor factorization, spectral density approximation, (matrix function) trace estimation, and solving linear systems and least squares problems.

My research focuses on the design and analysis of fast and theoretically justified (randomized) algorithms for core linear algebra tasks, with the overarching goal of developing tools to support the advancement of knowledge in current scientific applications. My approach includes (i) improving our theoretical understanding of algorithms already in use, (ii) making theoretical algorithms more suitable for real-world computers, and (iii) identifying domain problems for which no good algorithms exist but where RandNLA methods can be effective. This approach is benefited by my expertise in Krylov subspace methods (KSMs), a powerful class of algorithms that have traditionally been used for solving linear systems and eigenvalue problems, but are increasingly used for a broader range of tasks across the sciences.

As I outline below, this approach has led to a number of significant research contributions, including the first theoretical guarantees for widely-used algorithms like Stochastic Lanczos Quadrature (used in quantum physics for 30+ years without rigorous analysis), introduced new randomized estimators for quantum thermodynamics that expand the Pareto frontier of simulation accuracy/time, and shed new light on algorithms and inherent limits for operator learning in scientific machine learning. In the coming years, I will expand this program in order to operationalize RandNLA for widespread use in the computational sciences.

I currently work as an Applied Research Lead at JPMorganChase, where I have been instrumental in producing academic papers on quantum-inspired and randomized algorithms, developing high-performance implementations of state-of-the-art RandNLA algorithms, and supervising more junior researchers. While my current position enables me to work at forefront of applied computational math, *I am inherently drawn to the collaborative spirit and open inquiry of academic research, as well as the opportunity to teach, mentor, and work directly with students.*

My background well-suited for NYU Shanghai. In particular, *my expertise in RandNLA compliments the Department's existing strengths* in machine learning, data science, and theoretical computer science, while providing expertise in in this increasingly important area of computer science. At the same time, my knowledge of practical algorithm will be a valuable resource for researchers from across the university who may be looking to make use of advanced algorithms in their own research. Finally, my connections industry, the NYU/NYC research community, and the broader numerical linear algebra community, will help me to support graduate students within the department.

In what follows, I provide a taste of my research program. The included examples have been carefully curated to provide a sense of breadth of my research interests and the flavor of the results I produce. I also outline my vision for the future of my research at NYU Shanghai.

NEW UNDERSTANDING OF EXISTING ALGORITHMS

Often, very good algorithms for a given task already exist and are widely used in practice. However, a better theoretical understanding of such algorithms can result in more effective use. *My work provides new theoretical guarantees and insights into many widely used algorithms* including Stochastic Lanczos Quadrature (SLQ) for spectrum and spectral sum approximation [1–4], minibatch heavy-ball momentum for linear regression [5], the Lanczos method for tasks involving matrix functions

[6, 7], the full orthogonalization method (FOM) and block-conjugate gradient method for linear systems [8, 9], randomized block Krylov iteration (RBKI) for low-rank approximation [10], and the KernelSHAP algorithm for approximating Shapley values [11].

First bounds. My work often provides the first theoretical guarantees for widely used algorithms. For instance, a widely used algorithm for approximating the spectral density (also called density of states) of a $n \times n$ symmetric matrix $\mathbf{A}$ is Stochastic Lanczos Quadrature (SLQ) [12, 13]. SLQ has been around for decades, but a theoretical understanding of how certain parameters should be set a priori was missing. In [1] we prove *the first theoretical guarantees for SLQ for spectrum approximation*.

Theorem 1. Suppose $\varepsilon \gtrsim n^{-1/2}$. Then, using $O(\varepsilon^{-1})$ matrix-vector products to $\mathbf{A}$ (and lower order arithmetic), SLQ outputs an approximation to the spectral density of $\mathbf{A}$ with Wasserstein error at most $\varepsilon \|\mathbf{A}\|$. Moreover, this bound is sharp.

My work on algorithms for spectrum approximation also provides key conceptual insight to their mathematical and practical (e.g. in finite precision) behavior [2–4].

Practical insights. My work provides insight into real-world behavior of algorithms that help practitioners more effectively apply algorithms to the problems they are interested in. For example, the Lanczos method is often used to approximate the action of matrix functions on a vector. A special case of this algorithm is the Conjugate Gradient algorithm, whose behavior is well understood; in particular, many techniques for error estimation exist [14]. On the other hand, for more general functions, the behavior of the Lanczos method is much less understood. In [6], we introduce a general, intuitive, framework that allows the existing wealth of knowledge about CG to be transferred to the Lanczos method for arbitrary functions. This enables the efficient computation of highly accurate a posteriori bounds that are suitable for use as stopping criteria, even in finite precision arithmetic! My approach has prompted a number of follow-up works that generalize it to non-symmetric problems, and block and rational algorithms [15–17].

Exposition. My focus on developing conceptual understanding of algorithms extends beyond the development of new theorems and papers. In particular, *I am committed to creation of exposition that helps practitioners better understand the algorithms they use*, with the aim of supporting the advancement of basic science. Recent work in this direction includes (i) a monograph on the Lanczos algorithm for matrix functions for scientists [4] that provides a general introduction to these algorithms and disseminates key ideas about the stability of Lanczos-based methods, and (ii) an online book for advanced undergraduates that gives a hands-on introduction to RandNLA algorithms through a wide range of numerical experiments [18].

NEW ALGORITHMS FOR MODERN COMPUTING

In addition to analyzing existing algorithms, *I develop new, theoretically justified, algorithms designed to be fast on modern computers* (e.g. that can take advantage of available vectorization, GPUs, distributed memory parallelism etc.). These include algorithms for tasks such as estimating the spectral density of matrices [1, 3], traces and partial traces [22–24], linear systems [9, 25, 26], and regression problems [27].

Conceptual insights. I am particularly interested in developing algorithms that provide new conceptual insights that may be useful as more general algorithmic tools.

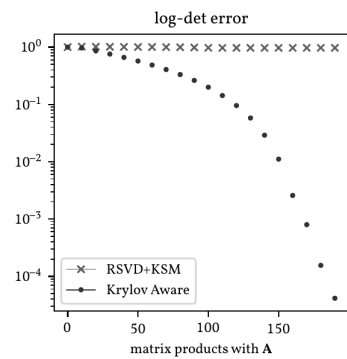


Fig. 1: Approximation error for our Krylov-aware algorithm vs. the previous state of the art RSVD+KSM approach [19–21] for estimating the log-determinant, an important quantity in data science.

For example, in [22], we study the problem of estimating the trace of a matrix function $\text{tr}(f(\mathbf{A}))$, an important task in a variety of domains, including data science and quantum physics. A number of recent algorithms for matrix function trace estimation aim to obtain variance-reduced trace estimators by first computing a low-rank approximation to $f(\mathbf{A})$ [19–21, 28]. To compute such an approximation, they combine black-box low-rank algorithms, such as the randomized singular value decomposition (RSVD) [29], with black box KSMs for approximating products with $f(\mathbf{A})$ [30]. In [22], we take a careful look into these black boxes in order to more carefully integrate the constituent subroutines. The resulting “Krylov-aware” *stochastic trace estimation algorithm commonly is orders of magnitude more accurate than previously state of the art algorithms*, despite a similar computational profile. Our *approach was recognized by the community as a key direction for further research* in a technical report on problems relating to matrix functions [31].

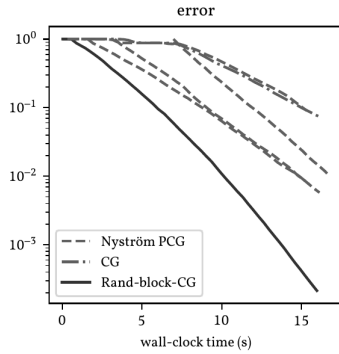


Fig. 2: The error as a function of runtime our randomized block-CG algorithm vs. the previous state of the art Nyström preconditioned CG approach [32] (with different choices of hyperparameters).

Another example of this type of work is my research on block KSMs for linear systems. On modern computer architectures, the cost of data movement is often much higher than the cost of arithmetic operations, meaning that applying a matrix (i) to a single vector or (ii) simultaneously to a block of vectors can have nearly the same cost. Interestingly, while block KSMs are widely used for tasks like eigenvalue problems/low-rank approximation, they are not as commonly used for solving linear systems. In [9], we reexamine this choice, by introducing a randomized variant of the block conjugate gradient method [33] for solving a single positive-definite linear system of equations. *Our algorithm provably outperforms preconditioned conjugate gradient with a broad-class of Nyström-based preconditioners* [32], without ever explicitly constructing a preconditioner.

Science oriented. I am particularly interested in extending techniques from RandNLA to develop algorithm with applications to the physical sciences. For example, I developed *the first randomized estimators for partial traces* [23, 24], a generalization of the trace

which is widely used in quantum physics. This enables the study of how quantum spin systems interact with one another at a scale/accuracy regime that was not previously accessible.

I have also worked on the development of practical algorithms for physics. For instance, by leveraging my background in orthogonal polynomials and KSMs, I developed a spectrum adaptive Python implementation of the Kernel Polynomial Method (KPM) is widely used in quantum chromodynamics and thermodynamics [34]. This implementation provides much better practical performance than existing methods by avoiding the need for a priori selection of hyperparameters. This algorithm is implemented in the spectral-density package for Python, available on PyPI.

Implementation. I have hands-on experience implementing and tuning algorithms for high-performance computing settings. For instance, in [27], we study the suitability of the sketch-and-precondition paradigm [35, 36] for solving large-scale linear regression problems on GPU devices. Our experiments are *run on 8 A100 GPUs*, and demonstrate the

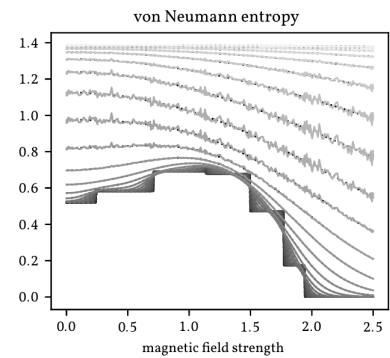


Fig. 3: von Neumann entropy of a subsystem as a function of temperature (indicated by color) and magnetic field strength. Our algorithm (provably) matches the analytic solution (dotted) closely.

effectiveness of our distributed implementation on near-terabyte scale problems. A key component of sketch-and-precondition is the choice of sketching matrix. While sparse sign sketches are theoretically powerful, they are difficult to efficiently generate in a simple way [37, 38]. Our work provides a carefully optimized high-level implementation of sparse sign sketches, which enables their use in practice, and may be of independent interest for large-scale software projects such as RandLAPACK. This project builds on the knowledge I developed while working on developing stable variants of the conjugate gradient algorithm for distributed memory clusters (which are implemented in PETSc) [25, 26].

BEYOND CLASSICAL NUMERICAL LINEAR ALGEBRA

I'm also interested in topics beyond classical numerical linear algebra such as scientific machine learning, a rapidly growing field which aims to use data-driven methods to learn the behavior of complex physical systems. An important task in scientific machine learning is operator learning, where we aim to learn an approximation of an operator (e.g. the solution operator to a given differential equation) from input/output data (e.g. forcing function/solution pairs). This approximation can then be used in downstream applications (e.g. to make predictions about future input data cheaply) [39, 40].

A widely studied version of this problem to learn an (unknown) $n \times n$ matrix $\mathbf{A}$ from matrix-vector queries $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ and $\mathbf{y} \mapsto \mathbf{A}^\top \mathbf{y}$ (such queries can e.g. correspond to physical experiments) [41–43] Typically the number queries m is much smaller than the dimension n , so to make the problem well-posed, we seek approximations from a structured hypothesis class (e.g. low-rank, sparse, hierarchical, etc.). Over the past several years, as part of an ongoing cross-disciplinary collaboration that I was central in organizing, I have studied the aforementioned problem for families such as hierarchical semi-separable (HSS) matrices [44], sparse matrices [45], and more [46].

As a concrete example, consider the class of hierarchical matrices are Hierarchically Off-Diagonal Low-Rank (HODLR) matrices. While there are some algorithms for learning an HODLR matrix [47–49], real-world matrices are not exactly HODLR. As such, there is *widespread concern* about whether such estimators might produce highly inaccurate approximations when applied to non-HODLR matrices [47, 50–53]. Notably, a recent SIAM Linear Algebra Best Paper asks whether a near-HODLR matrix can be approximated at nearly the same cost as required to learn an exactly HODLR matrix [54].

In [55] we provide *the first proof that we can learn a near-optimal HODLR approximation to an arbitrary matrix.*

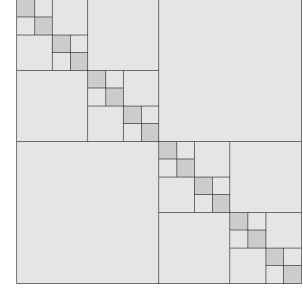


Fig. 4: Example of a HODLR matrix. Each off-diagonal block has rank $\leq k$.

Theorem 2. Using $m = O(k \log^4(n)/\epsilon^3)$ matrix-vector queries, we can obtain a rank- k HODLR approximation $\hat{\mathbf{A}}$ to $\mathbf{A}$ that is within a $(1 + \epsilon)$ factor of the best possible rank- k HODLR approximation to $\mathbf{A}$.

We also prove lower-bounds that show *no estimator for HODLR approximation can get the same guarantee with significantly less data*, regardless of how the data is generated. In order to prove Theorem 2, we develop a new perturbation bound for a broad class of low-rank approximation algorithms, including the widely known Randomized SVD [29]. Our work also provides key intuition about why certain low-rank approximation algorithms may be more susceptible to noise.

This example highlights the potential of this research direction. First, and in contrast to many tasks in NLA, this model of computation is amenable to lower-bounds on the number of input/output pairs needed to learn an approximation to $\mathbf{A}$. By studying such lower-bounds we can understand the limits of what can be learned from input/output data. This is relevant for algorithm design, as it provides insight into the types of families we use as hypothesis classes. Second, by applying existing tools from RandNLA to a new testbed of problems, we obtain new insights into core RandNLA algorithms which are of independent interest.

LOOKING FORWARD

My primary aim for the next decade is to operationalize RandNLA for widespread use on real-world computers, with the overarching goal of supporting the advancement of basic science. In particular, I will expand on the program described above to develop a better understanding of how to use the algorithms we already have, as well as to develop new algorithms for linear-algebra tasks. While I intend to continue working on a broad range of problems, I will place particular emphasis on the following thrusts:

Error bounds/estimates. I will develop a posteriori error bounds and estimates for RandNLA algorithms. Such techniques will simplify the practical usage of RandNLA algorithms, and are necessary for their widespread adoption by practitioners from other fields. Interestingly, while randomness is often advantageous for developing such estimates (e.g. using statistical techniques), there is a significant lack of study on obtaining such bounds.

Practically efficient algorithms. I am especially interested in algorithms that are designed for the way real computers work (e.g. cache hierarchies, parallelism, etc.). A promising area of study is (randomized) block Krylov methods, which are well-suited for modern computation due to their use of BLAS3. However, the behavior of block KSMs is far more complex than their single vector counterparts, necessitating new algorithmic and theoretical developments to unleash their full potential.

Software. An important aspect of my work in the near future will be the development of Scipy compatible open-source software for core linear algebraic tasks. I have already begun planning the organization and necessary methods for libraries on (i) RandNLA and (ii) KSMs for tasks involving matrix functions. I believe that both topics represent a remarkable gap in the study of practical algorithms and their actual implementation in software that can be easily used by practitioners. As such, both provide an opportunity for immediate impact.

The mentorship of students is an important aspect of my research program. My research program and my connections to industry and the NYU/NYC research community will help recruit and retain top PhD students. Likewise, my connections to faculty at Courant and Tandon will help ease the transition of NYUSH PhD students from their first year in New York to subsequent years in Shanghai, as well as help attract visiting students from the main NYU campus through initiatives like the Global Research Initiative fellowships. In addition, as I expand on in my teaching statement, I plan to include undergrads in my research program. I have been consistently successful in conceptualizing and managing projects that provide meaningful research experiences for undergrad students, and these projects regularly result in publications [9, 15, 24]. I'm excited for the strength of NYU Shanghai's undergrad students.

Within the first year and a half, I will begin submitting grants to obtain funding to support students and my research program. I have already started preparing materials for a grant to develop software for randomized KSMs. Upon joining NYU Shanghai, I would immediately begin familiarizing myself with the Chinese grant system and adapting my existing materials as needed.

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